The Research Program

A proof, prototyping, and experimentation roadmap
for the coordination results

Synthesized from three reviews, the provable-results portfolio,
the tooling survey, and the sheaf assessment — Document 3 of 4

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August 2026

What this consolidates. The prior portfolio enumerated eighteen results I judged provable end-to-end (tiers A/B/C). The three reviews then sharpened several, dissolved one, and added three genuinely new theorem candidates. The tooling survey fixed the formal substrate (Lean 4 + Mathlib for algebra/economics, Isabelle/HOL + AFP for noninterference, PRISM-games for inspection equilibria, Z3/cvc5/Gambit as the refutation layer). The sheaf assessment produced a commit-or-cut experiment. This document is the integrated program: what to prove, in what order, with which tool, gated by which experiment, and with an explicit *falsification-first* discipline so the program stays honest.

1 The discipline: refute before you prove

Adopt one rule across every result: **no bound is formalized until it survives an automated counterexample search.** Concretely, for each claimed inequality or equilibrium, first run a randomized parameter sweep through Z3/cvc5 (algebraic bounds), Gambit (finite-game equilibria), or PRISM-games (stochastic-game equilibria); only a claim with zero surviving counterexamples across the sweep is promoted to a Lean/Isabelle proof. This inverts the usual order and catches the errors the reviews caught by hand (the non-PD payoff matrix, the wrong δ threshold) *mechanically and early.* LLM provers (Goedel-Prover-V2 self-hosted, or the Aristotle API) sit *behind* the kernel as proof-search accelerators; accept only kernel-checked output, and human-review every Lean *statement* (autoformalization can produce a valid proof of the wrong theorem).

2 Tier A — days each (the legibility package + quick wins)

These are short proofs where the care is in the definitions. The first three assemble into one paper (Document 4, Paper 1).

A1. Split-Digest Theorem. *Lift: 1–2 days. Tool: Lean (order theory) + hand proof.* A single scoring head serves two readers iff their induced preference orders are comonotone; the successor reader (continuation value) and the operator reader (regret-if-ignored) are provably non-comonotone via two constructible items, so no shared head exists. The quantitative half: applying the information floor per reader, a *joint* zero-miss digest pays the sum of the two floors when the load-bearing sets are disjoint. *Gives:* converts the SLM-sidecar architecture from opinion to necessity; upgrades the original split-ranker theorem from example to characterization. *Review impact:* the rigor review independently argues “the operator digest and the context compaction must be the same provenance-preserving projection” — this theorem says *when* that is possible (comonotone readers) and when it is not, sharpening their claim.

A2. Deriving the regret head. *Lift: 1–2 days. Tool: Lean (real analysis) + hand proof.* regret = stakes × irreversibility × anomaly is not a design choice but the exact expected miss-cost *iff* anomaly is a calibrated posterior; the surfacing rule is then the classical likelihood-ratio criterion, and reputation r enters cleanly as a shift of the class-conditional danger probability. *Gives:* repairs the SDT error (Document 1, Change 4) constructively — the corrected Chapter I gets a derived optimal alerting policy with a calibration obligation, not a patched heuristic. *This is the single highest-value-per-day item*: it turns a flagged error into a theorem.

A3. ε-conservation for the release ledger. *Lift: 1 day. Tool: Lean or TLA$^+$/Apalache (state-machine invariant).* Induction on the append-only release transitions gives cumulative ε-spend $= \sum$ recorded releases, identical proof shape to bond conservation, plus imported DP composition. *Gives:* the Sealed Harbor's headline invariant; the structural rhyme (the harbor conserves one more scarce quantity through buckets) that makes the airlock auditable.

A4. Canary detection power + sequential leak testing. *Lift: 1 day. Tool: hand proof + numpy simulation.* $1 - \beta^k$ detection for a k-canary leak, plus a Wald SPRT giving expected time-to-detection. *Gives:* assurance-as-a-quantity on the leakage side; re-scopes the exfiltration bond to funding detection (breach damage being unbounded, per the product reviews' own $c = \infty$ admission).

A6. No-mint reputation inheritance. *Lift: 2–3 days. Tool: Lean (potential-function / supermartingale argument).* Per-outcome derivation budgets split across all derivations make total inherited prior $\leq$ total witnessed value, with distillation cycles safe. *Gives:* closes the skill-versioning cold-start gap safely and puts the product reviews' "Darwinian prompt Elo" on a sound footing (Document 2 hardening).

A7. The information-floor falsification experiment. *Lift: hours to first figure; runnable in-container now. Tool: numpy/scipy.* Sweep digest token budget through the floor $\log_2 \binom{N}{k} - \log_2 \binom{m}{k}$; overlay empirical miss rate; second panel corrupts a feature fraction ϕ to trace the mimicry/inflation attack; third panel measures the joint-digest penalty (A1) for disjoint vs. overlapping reader sets. *Gives:* the volume's first datum and best figure, and a genuine stress-test — if real digests beat the floor, the load-bearing formalization is wrong, and finding that out cheaply is the point.

3 Tier B — roughly a week each

B1. The digest–zoom Pareto frontier. *Lift: 1 week. Tool: Lean/hand for the convex program + numpy.* A two-constraint rate–distortion program $R(\delta, f)$ (false negatives priced by δ, false positives by open-budget f) with the original information floor as its worst-case corner; plus a *zoom-advantage theorem* quantifying what adaptive second-stage inspection (Hwang-style splitting inside the flagged set) buys over a flat digest. *Gives:* Chapter I's §I.6 grows from one bound into a small theory with an operator-time knob; the strongest standalone-paper candidate.

B2. The inspection tower. *Lift: 1–1.5 weeks. Tool: PRISM-games (equilibria) + Gambit (numeric) + Lean/Isabelle (closed form).* Stage result $\rho^\star = G/(dB)$; contraction under sealed sampling from q disjoint principal-cliques giving $\lambda = O(1/q)$, so finite bond capital certifies an infinite tower; and a declining audit schedule proving *reputation is amortized verification* ($O(1)$ lifetime audit spend per honest judge vs. $\Theta(T)$ flat). *Review synthesis (the crown jewel):* this is exactly where the two reviews' rate-the-raters proposals fuse — the product reviews' *model heterogeneity* supplies the disjoint cliques the contraction needs, and their *VRF honeypots* are the randomized audit realizing $\rho^\star$. B2 is the theorem behind both. Converts Conjecture III.11.1 into a parameterized result plus a measurement obligation (does the deployed judge pool satisfy sealed multi-clique sampling?).

B3. Hypervisor enforceability = Ramadge–Wonham controllability. *Lift: 1 week. Tool: Isabelle/Coq (meta-theorem) + Supremica (concrete synthesis).* Port Daddy's hypervisor can *regiment* policy P iff P is a prefix-closed safety language controllable w.r.t. the uncontrollable event set (model-internal steps, in-context content); everything else is detect-and-compensate. *Review synthesis:* this is the formal core of the product reviews' OP-2 ("regimentable iff a pure function of committed local state") stated correctly and generally, and it is the theorem the Sealed Harbor (C1) depends on. Yields a table re-grading every ledger row against the controllable set: force-push (network event) regimentable; "confident lie in a report" (semantic content) forever detect-only.

B4. A tractable deontic-conflict fragment. *Lift: 1 week. Tool: Z3/cvc5 (Horn + difference logic) + hand proof.* Consistency of a commitment language of ground Horn clauses + interval resource claims + difference constraints is polynomial and *incremental* (unit propagation + interval trees + negative-cycle detection); one step outside (disjunctive obligations) is NP-complete by a SAT reduction. *Review synthesis:* the rigor review's Alice/Bob/Carl chapter needs exactly this — "who decides what is a conflict" splits into mechanical detection (this fragment) and chartered resolution, and the schema designer gets a theorem stating which expressive conveniences forfeit tractability. Composes with B3: since acceptance is a mediated write, conflicting acceptances become *regimentable* pre-commit.

B5. Engine substitution / resurrection soundness. *Lift: 4–6 days. Tool: Lean (two-period lemons model).* Akerlof unraveling *inside one identity* (build reputation on a strong engine, spend on a weak one); closed by engine attestation, giving reputation on (principal, engine) pairs; resurrection soundness then preserves the sanction-respecting property across provider migration. *Gives:* the Phase-4 flagship demo ships with its own security theorem and names its own attack; one rule closes both engine-substitution and skill-versioning.

B6. The front-loaded probation theorem. *Lift: 3–5 days. Tool: Lean (exchange argument).* The optimal newcomer-ceiling schedule is maximally front-loaded — a probation *cliff*, not a gradual ramp — because the deterrence-to-honest-cost ratio $(\delta_f/\delta_h)^t$ strictly decreases in t. Mildly counterintuitive against deferred-compensation instincts, and the resolution is instructive. *Gives:* pins down *which* newcomer schedule repairs the (corrected) identity theorem, with a behavioral prediction a deployment can A/B against churn.

B7. Costly-escalation signaling. *Lift: 4–6 days. Tool: Lean (single-crossing) + Gambit.* A separating threshold equilibrium for the attention-queue alarm channel, with a computable tuning band $[\delta_{\min}, \delta_{\max}]$ for the dismissal debit. *Gives:* the "crying wolf is costly" mechanism gets existence + an operating band fit from measured validation rates, not vibes.

B8. The specialization boundary. *Lift: 3–4 days. Tool: Lean/numeric (Erlang-C).* Sole-responsibility roles beat pooling iff the skill-rate premium clears an explicit queueing threshold; includes the single-point-of-failure corner (M/M/1 with breakdowns) that quantifies the succession-rule requirement. *Gives:* the role-lease pattern (Phase 2) gets an honest boundary instead of an aesthetic.

B9. Context paging with an untrusted pin oracle. *Lift: 4–5 days. Tool: Lean/hand (competitive analysis).* Landlord/weighted-caching competitiveness on the unpinned region plus a linear ϕ-degradation term for a corrupted pin oracle — sharing one adversarial knob ϕ with the digest theory. *Gives:* the buffered-input layer (Phase 1–2) gets a competitive guarantee and the formal restatement of the effective-context trap.

4 Tier C — multi-week

C1. Sealed-Harbor noninterference via unwinding. *Lift: 2–3 weeks. Tool: Isabelle/HOL + AFP (Rushby intransitive-noninterference is already formalized there).* Domains $\{D, E, G_{\to E}, G_{\to D}, \text{pub}\}$

with an intransitive flow policy; discharge Rushby's three unwinding conditions per transition class; complete mediation (B3) discharges the "no other edges" premise; gate correctness is the named residual oracle. Possibilistic, honestly scoped (timing/termination channels out of model; the theorem bounds channels, not minds). *Review synthesis:* this is the rigor review's "formal claim worth attempting" (observational noninterference modulo declassification) made into a mechanized obligation, and it is the airlock's central promise. *Depends on B3.*

C2. Minimum viable take-rate. *Lift: 2 weeks. Tool: Lean (VCG deficit in a uniform-type model).* Closed-form expected VCG deficit for efficient bilateral trade $\Rightarrow$ the minimum platform fee that buys ex-post efficiency out of the Myerson–Satterthwaite corner; plus the scoped comparison that under common-value contamination (shared oracle) the same budget is better spent on disclosure than subsidy. *Gives:* Chapter 7's most-cited impossibility becomes a priced budget line.

C0 (new). The work-unit transition system. *Lift: 2–3 weeks. Tool: TLA$^+$/Apalache (bounded model check) $\to$ Isabelle (unbounded safety).* The rigor review's most productive formal object: one work unit plus principals, roles, capabilities, effects, claims, receipts, with transitions (propose, authorize, assign-role, request-effect, admit-effect, checkpoint, assert-claim, verify, contest, reassign, revoke, settle) and six safety properties (every admitted effect has an active grant/correct principal/role epoch/resource mapping; each idempotency key settles at most once; a verified completion has a path to acceptance policy and an admitted verifier result; no child capability more permissive than its parent; role reassignment invalidates stale epochs; every settlement is correctly funded). *Why it leads C-tier:* it is the formal spine of the re-spined volume (Document 1, Move 1) and the substrate every other theorem plugs into — prove its safety once and A-, B-tier results become properties *of* it rather than free-floating lemmas. Liveness is separated honestly (fair service, reachable approvers, bounded partitions).

5 The sheaf question: a bounded, decisive experiment

The sheaf-cohomology assessment reached a conditional verdict, and the decisive test is cheap. *Do this before writing a single sheaf sentence for the volume.* Build a hand-rolled numpy/scipy prototype (PySheaf cannot compute cohomology): gossip graph as a 1-complex with the peer-overlap cover, stalks as delta-vector embeddings of log states, coboundary δ, harmonic residuals via $\ker L_1$. Run honest baseline (must give H^1 = topological β_1 only) and single-equivocator cases on ring + partition topologies.

Verdict. **Commit-or-cut gate.** If dim H^1 (net of β_1) does not equal the number of injected equivocations, or if the *partition* regime yields no cohomology-only detection that pairwise digest comparison misses, **cut** the sheaf chapter to an appendix remark — cohomology would be renamed equivocation-detection. If both hold, the value-add is real (detection under partial visibility, where pairwise comparison is impossible but cocycle conditions over triangles still bind — structurally identical to Abramsky–Brandenburger contextuality), and the chapter commits with three theorem candidates (independent-equivocation counting, harmonic-support localization, consistency-radius = max prefix divergence). The consistency-radius result is low-risk and publishable regardless. *Honest note:* the abelianization gap (Carù's counterexamples) means vanishing H^1 over $\mathbb{R}$ certifies consistency only up to that gap; non-vanishing is a sufficient alarm, never treat vanishing as an all-clear without a pairwise sweep.

6 Experiments that matter (adversarial differential, not demos)

The rigor review's experiment list is the right empirical program; pair each with the theorem it tests.

- *Raw chat vs. typed acts* (duplicate work, hidden commitments, intervention time) — tests the messaging-layer read bound and the value of deontic typing.
- *Digest vs. claim-graph* (decision accuracy, false belief, time, tokens, source coverage) — tests A1/A2/B1.
- *Sidecar attacks* (injection, poisoned evidence, omission, conflicting sources, summary recursion) — tests the projection-sidecar soundness boundary and the ϕ-parameter shared with B9.
- *Hook mode vs. confined mode* (bypass success across fs/git/net/credentials/logs/child processes) — tests B3 and the assurance-level claims empirically; this is also the red-team gate for the Phase-2 "Confined (level 4)" claim.
- *Cross-provider resurrection* (time to useful next action, duplicated effects, obligation loss, final verifier outcome) — tests B5 and the capsule fidelity metric.
- *Role failover under kill/partition* (stale effects, mailbox continuity, SLO recovery) — tests the exclusive-role epoch safety and the CAP trade.
- *Blind review fleet* (seeded real defects, false-block rate, joint-miss correlation, post-merge rollback) — tests the independence assumption directly; measures the pairwise/higher-order joint-miss rates that the $(1-d)^k$ story needs.
- *Lookout/Unspider* (future incidents only; precision, recall, Brier calibration, warning lead time) — tests prospective-assurance soundness within the declared abstract domain.
- *Sealed-Harbor red team* (canary exfiltration through network, logs, errors, filenames, length, timing, CPU load, receipt status) — tests A4 and the C1 side-channel scope honestly.

7 Batching and the critical path

- **Week 1:** A2 + A1 + A7 (the legibility package: the derived regret head, the split-digest characterization, the falsification figure). Ship-shape as Paper 1's core.
- **Week 2:** B3 (hypervisor controllability) — highest leverage; re-grades the ledger, earns the Phase-2 product claim, and unblocks C1.
- **Week 3:** B2 (inspection tower) — the rate-the-raters synthesis; converts the central conjecture.
- **Week 4:** B1 (the frontier) — Paper 1's theoretical heart.
- **Parallel track (whenever):** the sheaf commit-or-cut experiment (2–3 days, decisive); A3/A4 (Sealed-Harbor invariants, cheap); A6 (skill inheritance).
- **The long arcs:** C0 (work-unit transition system — start early, it is the spine everything plugs into), then C1 (Sealed-Harbor noninterference, gated on B3), then C2.
- **The Chapter III pair:** B5 + B6 together (engine substitution + probation cliff).
- **The Alice/Bob/Carl core:** B4 (tractable conflict fragment).

What I still decline to claim (in the ledger's spirit): gossip convergence under real partitions (an empirical distribution, correctly graded by the volume); cross-operator attestation closed as a *proof* (a hardware/PKI deployment problem — specify the TEE / key-transparency evaluation, do not prove the gap shut); folk-theorem cartel bounds at federation (provable only in monitoring models stylized past usefulness); and the sheaf program *beyond* the committed experiment. Every result touching deployment parameters (d, β, ϕ, calibration maps, validation rates) is a theorem *conditional on measurement* — the instrumentation is the co-requisite, not the afterthought.